06. Change of Variables

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In General, if we have a random variable $X \in \mathbb{R}$ ← real valued

with distribution $F(x)$

Then what is the distribution of $Y = \phi(X)$

for some Faction ϕ?

Similar Problem, what are the CDF and PDF of Y, $F_Y(y)$, $f_Y(y)$?

* The probability density Function is not really a Function, it is a derivative

Example: X is a random Variable

$Y = 2X$

- CDF: $F_Y(t) = P(Y \le t)$

$= P(2X \le t)$

$= P(X \le t/2) = F_X(t/2)$

- PDF: $f_Y(t) = \frac{d}{dt} F_Y(t) = \frac{d}{dt} F_X(t/2) = \frac{1}{2} f_X(t/2)$ $\quad \frac{d}{dt}(t/2)$

$Y = \phi(X) \Rightarrow X = \phi^{-1}(Y)$

$= 2X \qquad = Y/2$

- $F_Y(t) = F_X(\phi^{-1}(t))$
- $f_Y(t) = f_X(\phi^{-1}(t)) \cdot \frac{d}{dt}\phi^{-1}(t)$

$F_X(t) = P(X \le t) = \int_{-\infty}^{t} f_X(t)\,dt$

Don't Do this: $Y = 2X$ then assume $f_Y(t) = f_X(2t)$

$= 2 f_X(t)$

(all) →

(all wrong) $\longrightarrow$ $= 2 f_X(t)$

$= f_X(t/2)$

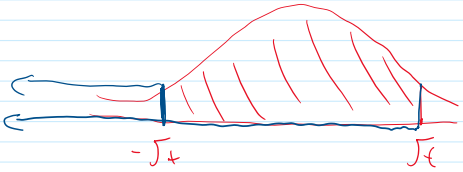
Example: Chi-squared Distribution

- Say $Z \sim N(0,1)$, $W = Z^2$ then $W \sim \chi^2(1)$
- Find the pdf of $W = Z^2$

$(t \geq 0)$

$$F_W(t) = P(W \leq t) = P(Z^2 \leq t) = P(|Z| \leq \sqrt{t})$$

$$= P(Z \leq \sqrt{t}) - P(Z \leq -\sqrt{t})$$



$$= \int_{-\infty}^{\sqrt{t}} f_Z(t)\,dt - \int_{-\infty}^{-\sqrt{t}} f_Z(t)\,dt = \int_{-\sqrt{t}}^{\sqrt{t}} f_Z(t)\,dt$$

$$= F_Z(\sqrt{t}) - F_Z(-\sqrt{t})$$

$$\frac{d}{dt} F_W(t) = \frac{d}{dt}\left(F_Z(\sqrt{t}) - F_Z(-\sqrt{t})\right)$$

$$f_Z(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$$

$$= \frac{1}{2\sqrt{t}} f_Z(\sqrt{t}) + \frac{1}{2\sqrt{t}} f_Z(-\sqrt{t})$$

$$= \frac{1}{2\sqrt{t}}\left[\frac{1}{\sqrt{2\pi}} e^{-\sqrt{t}^2/2} + \frac{1}{\sqrt{2\pi}} e^{-(-\sqrt{t})^2/2}\right]$$

$$= \frac{1}{\sqrt{2\pi}} \boxed{\frac{1}{\sqrt{t}} e^{-t/2}}$$

(wrong Answer

$$f_W(t) = \frac{1}{\sqrt{2\pi}} e^{-\sqrt{t}^2/2}$$
)

Short Version: Multiply by derivative of the transform!

Next time: Box-Muller transform

If $U \sim Unif[0,1]$, $V \sim Exp(1)$, U, V independent

Then $$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \sqrt{2V}\cos(2\pi U) \\ \sqrt{2V}\sin(2\pi U) \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right)$$

* Change of Variables in two Dimensions

Now Multiply by the ~~Derivative~~ Jacobian